

Name _____ Period _____

Skill 32.1 Exercise 1

Refer to the code segment below,

```
import numpy as np
coin = ["H", "T"]
flips = np.random.choice(coin, size=20, replace=True)
num_heads = np.sum(flips == "H")
print(flips)
print("Heads:", num_heads)
```

What data type is stored in flips?

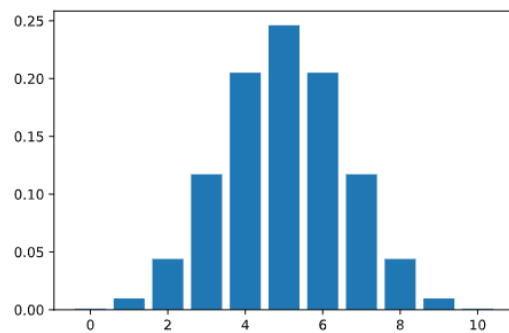
What are the only possible values?

How many flips are simulated?

If you set replace=False, would this code work with size=20? Why or why not?

Skill 32.2 Exercise 1

The graph below shows the PMF for flipping a fair coin 10 times.



X = number of heads in 10 coin flips

What does X represent?

What is the probability of getting exactly 5 heads? 0 heads? 10 heads?

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Skill 32.3 Exercise 1

A biased coin lands on heads with a probability $p=0.60$. You flip it $n=10$ times. Write a loop to calculate the probability of getting 0 to 10 heads using the *pmf* method below.

```
stats.binom.pmf(k, n, p)
```

Which outcome is most likely now: 5, 6, or 7 heads? Explain.

Skill 32.4 Exercise 1

A six-sided die has a $1/6$ probability of rolling a number 6. Use the *pmf* method below to show how you could calculate the probability of rolling between 1 and 3 sixes when you roll a fair die 8 times.

```
stats.binom.pmf(k, n, p)
```

Skill 32.5 Exercise 1

Use the cumulative distribution function below to show how you could calculate the following,

```
stats.binom.cdf(k, n, p)
```

The probability of observing 3 or fewer heads from 10 fair coin flips

The probability of observing more than 5 heads from 10 fair coin flips

The probability of observing between 2 (inclusive) and 5 (inclusive) heads from 10 fair coin flips

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Skill 32.6 Exercise 1

The weather in the Galapagos islands follows a Normal distribution with a mean of 20 degrees Celcius and a standard deviation of 3 degrees.

Write code to calculate the probability that the weather on a randomly selected day will be between 18 to 25 degrees Celcius using the *norm.cdf()* method.

```
scipy.stats.norm.cdf(x, loc, scale)
```

- x: the value you want
- loc: the mean (default = 0)
- scale: the standard deviation (default 1)

Using the same information about the Galapagos Islands, write code to calculate the probability that the weather on a randomly selected day will be greater than 24 degrees Celsius.